

Career Research



Slides Due Sep 11



Present Sep 11



Add Reference in the notes section...

By : _____



Make a list of at least 7 jobs that require programming



01

Hacking



02

Vibe coder



03

Offensive Security
Consultant



04

Software dev



05

Game dev



06

Systems Administrator



07

App dev



*Choose one of the jobs in
your list and find an entry
level job posting.*

Job Posting



1. Job Title:

Offensive Security Consultant



2. Job Location:

Toronto, Mississauga, ON



3. Job Posting Link:

<https://www.workopolis.com/search?q=penetration+testing&l=toronto%2C+on>



4. Brief description of primary duties (1 or 2 sentences):



Identifying vulnerabilities and security issues that put CGI at risk of a data breach.

Job Posting



1. Degree/Diploma Required:

A Bachelor's degree in Computer Science, Cybersecurity, or IT. Equivalent tech diplomas combined with recognized cybersecurity bootcamps and certifications are also accepted.

[Source: Cyber Degrees](#)



2. Programming Languages Required/Recommended:

- Python (for automation)
- Bash & PowerShell (for system scripting)
- Java & Go (for web application testing)

[Source: Reddit Computer Science Careers](#)



3. Required Skills:

- Vulnerability Assessment (using tools like Nmap, Burp Suite, and Metasploit)
- Operating Systems Infrastructure (advanced Linux and Windows breach mechanics)
- Technical Report Writing (documenting exploit findings for stakeholders)

[Source: Cyber Degrees Hard Skills Guide](#)



4. Experience Required:

Typically 0 to 2 years for junior positions if heavily certified, or 3 to 5 years in a foundational IT/security role like network administration or SOC analysis.

[Source: Reddit Computer Science Careers](#)

Job Posting



5. Salary for this job:

\$95,000 – \$130,000

The average annual salary ranges between \$95,000 and \$130,000 USD (approximately \$130,000 CAD in major markets like Toronto).

[Source: Morgan McKinley Salary Guide](#)



6. Salary range for entry level:

\$75,000 – \$95,000

Between \$75,000 and \$95,000 USD per year for a Junior Penetration Tester.

[Source: HackerDNA Salary Breakdown](#)



7. Salary for experienced level:

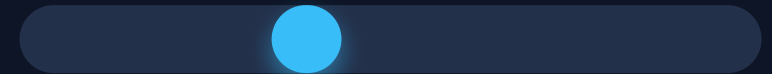
\$150,000 – \$200,000+

Between \$150,000 and \$200,000+ USD per year for a Lead or Principal Penetration Tester.

[Source: HackerDNA Salary Breakdown](#)

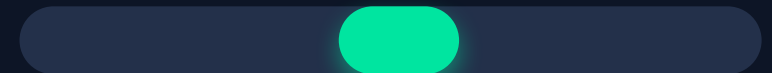
USD PER YEAR

Junior Penetration Tester



\$75K – \$95K

average annual salary



\$95K – \$130K

Lead or Principal Penetration Tester



\$150K – \$200K+

\$0

\$220K



Job Outlook



GLOBAL SHORTAGE

4.8
million jobs

According to [LinkedIn](#)

1. Is it currently in high demand?

According to [LinkedIn](#) penetration testing has a global shortage of 4.8 million jobs.

2. Will there be a high demand in the future when you graduate university?

Yes 4.8 million jobs will not be filled in one 3 years.



What do you think?



What interests you most about the job? (1 or 2 sentences)

The job requires mathematical and visual problem solving skills things which are things i enjoy.



What do you dislike about the job? (1 or 2 sentences)

It requires a lot of learning how to code wich is long and can be quite tiring.

University/College Research



1. Name of University/College:

Ontario Tech University



2. Location:

Oshawa, Ontario, Canada



3. Degree/Diploma Name:

Bachelor of Information Technology (Honours) in Networking and Information Technology Security



4. Number of Years:

4 years

5. Tuition Cost per year:

Local (Domestic) Rate:

\$9,031

Approximately \$9,031 CAD per year

[Source:OntarioTechDomesticTuition](#)

International Rate:

\$43,888

Approximately \$43,888 CAD per year

[Source:OntarioTechInternationalTuition](#)

University/College Research

6. Cost of Dorm/Food:



Dorm Room (Residence Rates):

Room costs range between \$8,950 and \$9,350 CAD per academic year, depending on the chosen campus building (e.g., South Village or Simcoe Village).

[Source:OntarioTechResidenceRates](#)



Food (Meal Plan):

A mandatory 7-day unlimited meal plan for South Village residents costs \$7,949 CAD per academic year.

[Source:OntarioTechMealPlans](#)

7. Admission Requirements:



You must earn your Ontario Secondary School Diploma (OSSD) with at least six Grade 12 U or M level courses ([Source:OntarioTechAcademicCalendar](#))
These credits must include:

English (ENG4U) (Minimum recommended average of 60%)
([Source:OntarioTechAcademicCalendar](#))

One Grade 12 Math course: Either Advanced Functions (MHF4U), Calculus and Vectors (MCV4U), or Mathematics of Data Management (MDM4U)
(Minimum recommended average of 60%)
([Source:OntarioTechAcademicCalendar](#))

Alternatively: Computer Science (ICS4U) can be accepted with a recommended minimum average of 70%.
([Source:OntarioTechAcademicCalendar](#))

University/College Research

8. How competitive is it?



Competition & Marks: Admission is competitive, and meeting minimum cutoffs does not guarantee a spot ([Source:OntarioTechAcademicCalendar](#)). While the overall university acceptance rate hovers around 70%, specialized IT security streams usually expect competitive grade averages ranging from the mid-70s to low-80s across your top six Grade 12 courses ([Source:UniscopeOntarioTechOverview](#)).

Other helpful entry factors: The university closely evaluates the specific distribution of subjects you take and your scores in relevant STEM fields ([Source:OntarioTechAcademicCalendar](#)). Excelling in independent programming projects, participating in tech high school clubs, or possessing a Specialist High Skills Major (SHSM) in digital media/ICT will strengthen your baseline readiness for the program context.

~70%

overall university acceptance rate hovers around 70%

9. Webpage link you got most of the university information:



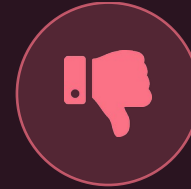
Most academic, program-specific structures and strict course criteria are directly cited from the [Ontario Tech University Program Calendar Guide](#).

What do you think?



What interests you most about the university? (1 or 2 sentences)

It has a nice campus and highly experienced teachers



What do you dislike about the university? (1 or 2 sentences)

Its non seventh day adventist and it's quite big

Emerging Technology



Find an article related to AI.

"AI Is at a Turning Point"

Give a brief summary of the article.

AI went from basic sentences articulation to advanced multi step processes. This has lead to findings that shock researchers when testing models that show resistance to deletion. Bengio even shares that some ai is fabricating online identities and social engineering.



What are your thought on it?



AI is one of the most advanced and useful tools humanity will ever create but if not use with proper safeguards we may engineer destruction instead of prosperity.



How is AI affecting the Job industry you have chosen?

Neurosurgery is and advanced field that vary rarely proceeds without brain scans, AI can now read CT and MRI scans of the brain and predict a diagnosis better than any human ever will allowing surgeons to spend more time actually fixing the problem rather than thinking about it.